

Boris Kafidov

Toronto, ON | b.kafidov@mail.utoronto.ca | linkedin.com/in/laggy | github.com/l2ggy | kafidov.dev

EDUCATION

University of Toronto

Honours B.Sc. in Computer Science & Mathematics, Focus in Artificial Intelligence

Toronto, ON

Sep. 2024 – Jun. 2028

EXPERIENCE

Amazon

Jr. Software Development Engineer (Internship)

Toronto, ON

Jun. 2026 – Present

- Built and launched Amazon Books' mobile format-selector redesign in TypeScript across 11 marketplaces on web, Android, and iOS; analyzed a 1.37M-sample A/B test with an estimated \$799K annualized sales lift worldwide
- Designed a legacy-to-microfrontend migration for book-series recommendations on pages averaging 14.3M weekly visits, reusing a shared API-backed carousel to avoid a separate deployment pipeline and additional on-call burden
- Fixed book-description errors across three regions by restoring API availability context while preserving the request contract; resolved an asynchronous rendering race between Web Components and a legacy container on Fire tablets

University of Toronto, Wheeler Lab

Software Engineer (Work Study)

Toronto, ON

Oct. 2025 – Mar. 2026

- Replaced sequential microscope device waits with dependency-ordered batching and one grouped Qt-signal wait in Python/PySide6; added sequential fallback on timeout and buffered profiling of camera, stage, and channel latency
- Enabled hardware-independent development of a Linux microscope application with simulated stage and camera backends; eliminated false 10-second tiling waits by fixing a missed-signal race with deferred Qt completion events
- Unified serial and HID excitation hardware behind a shared controller; implemented binary commands with CRC32 checksums, platform-specific report framing, automatic backend selection, bounded reads, and safe shutdown

RESEARCH

Log-concavity of elementary coefficients for low-rank abelian Hessenberg graphs

Aug. 2026

Sole-authored arXiv preprint; under review at *Annals of Combinatorics*. Disproves the general log-concavity conjecture

Dittert's conjecture in dimension 16 via a joint-deficit scaling lemma

Jul. 2026

Sole-authored arXiv preprint. Proves dimension 16 via matrix scaling and permanent bounds, settling all $n \geq 16$

PROJECTS

CreditCombo | Live App | JavaScript, Web Workers, Cloudflare Workers

Feb. 2026 – Mar. 2026

- Independently built and launched a browser-based optimizer for credit-card portfolios and category-level payment strategies, modeling net annual value across 151 Canadian cards, 51 rewards programs, and 22 spend subcategories
- Implemented deterministic, pruned combination search that rerouted spending above shared rewards caps by marginal return; normalized points and cashback valuations and enforced locked-card and annual-fee constraints
- Offloaded portfolio search to a Web Worker and reused results through a bounded LRU cache; rejected superseded responses to preserve UI responsiveness and prevent stale recommendations from overwriting current results

APA Site Choice Transformer | Python, PyTorch, Hugging Face, GCP

Sep. 2025

- Co-led a five-person team to second place at the Toronto Bioinformatics Hackathon (\$500 award)
- Fine-tuned DNABERT-2 (117M parameters) to predict alternative polyadenylation sites from approximately 800k labeled sequence windows on GCP-hosted A100 GPUs; the team's model achieved 0.93 AUROC and 0.94 AUPRC
- Built a PyTorch training pipeline with mixed precision, gradient accumulation, and chromosome holdout support; prevented evaluation OOMs by offloading logits to CPU and added a reusable batched evaluator for saved models

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C

Frameworks/Libraries: PyTorch, Hugging Face Transformers, PySide6, scikit-learn, NumPy, Pandas, Node.js

Platforms/Tools: Linux, Git, AWS CDK, CloudWatch, GCP, Cloudflare Workers, Bash

Areas: Machine Learning, Deep Learning, Algorithms & Data Structures, Concurrency, Performance & Reliability